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Sheet feeding apparatus and image forming apparatus

Abstract

A sheet feeding apparatus includes a first supporting portion movable between a closed position and an open position supporting a sheet, an interlocking portion to move the first supporting portion from the closed position to the open position when a cover is opened, and a second supporting portion. A feed portion feeds a sheet supported by the first and second supporting portions in a sheet feeding direction. A moving angle of the first supporting portion moving from the closed position to the open position is smaller than a moving angle of the cover moving from the closed position to the open position, and the first supporting portion includes an upstream end portion and a downstream end portion. In a state where the cover and first supporting portion are open, the upstream end portion is further from an exterior surface of the cover than the downstream end portion.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7748691	12/2009	Mizuguchi	271/171	G03G 15/6514
9604806	12/2016	Okura	N/A	B65H 1/04
9665053	12/2016	Mori	N/A	G03G 15/6514
10053310	12/2017	Nishiyama et al.	N/A	N/A
11286124	12/2021	Kannari	N/A	B65H 1/04
11795017	12/2022	Nishiyama	N/A	B65H 1/04
2009/0267283	12/2008	Mizuguchi	N/A	N/A
2016/0332832	12/2015	Okura	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H10-245141	12/1997	JP	N/A
2007-153455	12/2006	JP	N/A
2009-263078	12/2008	JP	N/A
2016216155	12/2015	JP	N/A

OTHER PUBLICATIONS

Tomoya Tateishi et al., U.S. Appl. No. 17/496,856, filed Oct. 8, 2021, Art Unit 1743. cited by applicant

Japanese Office Action dated Oct. 1, 2024, in related Japanese Patent Application No. 2020-205494. cited by applicant

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a sheet feeding apparatus configured to feed a sheet and to an image forming apparatus including the same.

Description of the Related Art

(2) Hitherto, a printer including a sheet feeding unit configured to feed a sheet loaded in a manual sheet feed portion has been proposed by Japanese Patent Application Laid-open No. 2016-216155 for example. The manual feed tray includes a cover openably and closably supported by an apparatus body of the printer, a base tray and an auxiliary tray supported on an upper surface of the cover and an interlock mechanism configured to move the base tray in an interlock manner with opening and closing operations of the cover. The auxiliary tray moves from a storage position to a lifting position along with the opening operation of the cover.

(3) According to Japanese Patent Application Laid-open No. 2016-216155, the auxiliary tray is configured to move from the storage position to the lifting position by being brought into pressure contact with a hook portion of the base tray.

SUMMARY OF THE INVENTION

(4) According to an aspect of the present invention, a sheet feeding apparatus includes an apparatus body, a cover openably and closably supported between a first closed position and a first open position with respect to the apparatus body, a first supporting portion movable between a second closed position where the first supporting portion is accommodated in the apparatus body and a second open position where the first supporting portion is capable of supporting a sheet, an interlocking portion configured to move the first supporting portion from the second closed position to the second open position in a case where the cover is opened from the first closed position to the first open position, a second supporting portion drawably supported by the first supporting portion and capable of supporting the sheet together with the first supporting portion, and a feed portion configured to feed the sheet supported by the first and second supporting portions. A moving angle of the first supporting portion in moving from the second closed position to the second open position is smaller than a moving angle of the cover in moving from the first closed position to the first open position.

(5) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram illustrating an overall configuration of a printer according to a present exemplary embodiment.

(2) FIG. 2 is a perspective view illustrating a manual sheet feed portion.

(3) FIG. 3 is a perspective view illustrating a front cover.

(4) FIG. 4A is a perspective view illustrating a link arm seen from one direction.

(5) FIG. 4B is a perspective view illustrating the link arm seen from another direction.

(6) FIG. 5A is a perspective view illustrating a manual feed tray seen from an upper surface side.

(7) FIG. 5B is a perspective view illustrating the manual feed tray seen from a bottom surface side.

(8) FIG. 6 is a perspective view illustrating an extension tray.

(9) FIG. 7 is a section view illustrating the manual sheet feed portion in a case where the front cover is positioned at a closed position.

(10) FIG. 8 is a section view illustrating the manual sheet feed portion in a case where the front cover is located at an open position.

(11) FIG. 9 is a section view illustrating a state in which the extension tray is drawn out of the

manual feed tray.

(12) FIG. **10** is a section view illustrating a state in which the front cover is closed while the extension tray is drawn out.

DESCRIPTION OF THE EMBODIMENTS

(13) Overall Configuration

(14) A printer **1** serving as an image forming apparatus of the present exemplary embodiment is an electro-photographic type full-color laser beam printer. As illustrated in FIG. **1**, the printer **1** includes an image forming unit **100** configured to form an image on a sheet S, a cassette sheet feed portion **300**, a fixing portion **400** and a manual sheet feed portion **500**. The image forming unit **100** includes four process cartridges **110Y**, **110M**, **110C** and **110K** respectively configured to form four color toner images of yellow (Y), magenta (M), cyan (C) and black (K) and a scanner unit **120**.

(15) Note that the four process cartridges **110Y**, **110M**, **110C** and **110K** have the same configuration except the colors of images to be formed are different. Therefore, the configuration and an image forming process of only the process cartridge **110Y** will be described below and descriptions of the process cartridges **110M**, **110C** and **110K** will be omitted.

(16) The process cartridge **110Y** includes a photosensitive drum **111**, a charging roller **113** and a developing roller **114**. The photosensitive drum **111** is constructed by applying an organic photoconductive layer on a circumference of an aluminum cylinder and is rotated by a driving motor not illustrated. Note that a photoconductive belt may be used instead of the photosensitive drum **111**. The image forming unit **100** includes an intermediate transfer belt **201** wound around a driving roller **105**, a tension roller **121** and others. The intermediate transfer belt **201** is rotated by the driving roller **105** clockwise in FIG. **1**. Provided inside of the intermediate transfer belt **201** are primary transfer rollers **115Y**, **115M**, **115C** and **115K**.

(17) The fixing portion **400** includes a fixing film **401** heated by a heater not illustrated and a pressure roller **402** in pressure contact with the fixing film **401**. The cassette sheet feed portion **300** is provided at a lower part of the printer **1** and includes a sheet feed cassette **301** drawable from and attachable to a printer body **1A** serving as an apparatus body.

(18) Next, an image forming operation of the printer **1** thus constructed will be described. As an image signal is inputted to the scanner unit **120** from a personal computer or the like not illustrated, the scanner unit **120** irradiates the photosensitive drum **11** of the process cartridge **110Y** with a laser beam corresponding to the image signal.

(19) At this time, a surface of the photosensitive drum **111** has been homogeneously charged with a predetermined polarity and potential by the charging roller **113** in advance and an electrostatic latent image is formed on the surface of the photosensitive drum **111** by the laser beam irradiated from the scanner unit **120**. The electrostatic latent image formed on the photosensitive drum **111** is developed by the developing roller **114** to be a toner image of yellow (Y) on the photosensitive drum **111**.

(20) In the same manner, the respective photosensitive drums of the process cartridges **111M**, **111C** and **111K** are also irradiated by the scanner unit **120** with the laser beams, and toner images of magenta (M), cyan (C) and black (K) are formed on the photosensitive drums, respectively. The respective toner images formed on the respective photosensitive drums are then transferred by primary transfer rollers **115Y**, **115M**, **115C** and **115K** onto the intermediate transfer belt **201** and are conveyed to a secondary transfer roller **203** by the intermediate transfer belt **201** rotated by the driving roller **105**.

(21) Note that the image forming processes of the respective colors are performed with a timing of superimposing on an upstream toner image primarily transferred onto the intermediate transfer belt **201**. Toner left on the surface of the photosensitive drum **111** after transferring each color toner image onto the intermediate transfer belt **201** is removed by a cleaning unit not illustrated.

(22) In parallel with the image forming process, a sheet S accommodated in the sheet feed cassette **301** of the cassette sheet feed portion **300** or a sheet supplied from a manual sheet feed portion **500**

is fed by a pickup roller **311**. The sheet S fed by the pickup roller **311** serving as a feed portion is separated one by one at a separation nip formed by a feed roller **312** and a separation roller **313**.

(23) A torque limiter not illustrated is attached to the separation roller **313** such that the separation roller **313** does not rotate in a case where two more sheets enter the separation nip and such that the separation roller **313** is rotated by the feed roller **312** in a case where one sheet enters the separation nip. Note that a so-called retard roller system of inputting a rotational drive in a direction opposite to a sheet feeding direction may be adopted instead of the separation roller **313**.

(24) The sheet S separated one by one by the separation nip is then conveyed to a registration roller pair **321**. The registration roller pair **321** corrects a skew of the sheet S and conveys the sheet S with a predetermined timing synchronized with the image forming process. Then, a full-color toner image on the intermediate transfer belt **201** is transferred onto the sheet S by a secondary transfer bias applied to the secondary transfer roller **203**.

(25) The sheet S onto which the toner image has been transferred is then conveyed to the fixing portion **400** to undergo predetermined heat and pressure by the fixing film **401** and the pressure roller **402** of the fixing portion **400** to melt and secure, i.e., to fix, the toner onto the sheet S. The sheet S which has passed through the fixing portion **400** is discharged onto a discharge tray **404** by a discharge roller pair **403**.

(26) Manual Feed Portion

(27) Next, a configuration of the manual sheet feed portion **500** serving as a sheet feeding apparatus will be described with reference to FIGS. **1** through **6**. As illustrated in FIGS. **1** and **2**, the manual sheet feed portion **500** includes a front cover **350**, a manual feed tray **351**, an extension tray **352**, link arms **353L** and **353R**, a sheet feed unit **354** and others. Parts configured to support at least the link arms **353L** and **353R** and parts configured to support the front cover **350** of the printer body **1A** are also included in the manual sheet feed portion **500** in the present exemplary embodiment.

(28) The front cover **350** serving as a cover of the manual sheet feed portion **500** is openably and closably supported to the printer body **1A** centering on an opening and closing shaft **355** between a closed position, i.e., a first closed position as illustrated in FIG. **7** and an open position, i.e., a first open position as illustrated in FIG. **8**. The front cover **350** is positioned at the closed position when no sheet is fed in the manual sheet feed portion **500** and composes a part of an exterior face of the printer **1** at this time. The front cover **350** is positioned at the open position when a sheet is to be fed in the manual sheet feed portion **500**.

(29) A user loads a sheet on a manual feed tray **351** attached to the front cover **350** while positioning the front cover **350** at the open position. In a case where a sheet to be fed by the manual sheet feed portion **500** is a long-size sheet for example, an extension tray **352** described later may be drawn out of the manual feed tray **351** to support the long-size sheet by the manual feed tray **351** and the extension tray **352**.

(30) The sheet feed unit **354** includes a pickup arm **354a** pivotably supported by the printer body **1A** centering on a feed roller shaft **357a**. The sheet feed unit **354** also includes a pickup roller **356** rotatably supported by the pickup arm **354a** and a feed roller **357**. As a job of feeding a sheet from the manual sheet feed portion **500** is inputted in a state in which the sheet is stacked in the manual feed tray **351** of the manual sheet feed portion **500**, the pickup arm **354a** pivots such that the pickup roller **356** is lowered.

(31) Then, the sheet is fed as the pickup roller **356** rotates in a state in which the pickup roller **356** is in contact with the sheet on the manual feed tray **351**. The sheet fed by the pickup roller **356** is conveyed toward the registration roller pair **321** by the feed roller **357** and conveyance roller pairs **359**, **360** and **361**.

(32) As illustrated in FIG. **3**, the front cover **350** includes an inner surface **350f** facing to the printer body **1A** at the closed position and ribs **350c**, **350g** and **350h** provided along outer edge of the inner surface **350f** and extending in a direction of separating from the exterior face **350m**. These ribs

350c, **350g** and **350h** are provided so as to rise up from the inner surface **350f** to improve stiffness of the front cover **350**.

(33) Provided also inside of the ribs **350c**, **350g** and **350h** of the front cover **350** is a pair of tray supporting portions **350kL** and **350kR** formed so as to rise up from the inner surface **350f**. A first guide rail **350aL**, a second guide rail **350bL** and a restriction rib **350dL** are formed on the tray supporting portion **350kL**. A first guide rail **350aR**, a second guide rail **350bR** and a restriction rib **350dR** are formed on the tray supporting portion **350kR**. The first guide rails **350aL** and **350aR** are disposed respectively downstream of the second guide rails **350bL** and **350bR** in terms of the sheet feeding direction FD. The restriction ribs **350dL** and **350dR** are formed so as to extend inside in the width direction W at upper end portions of the tray supporting portions **350kL** and **350kR**.

(34) Next, while the link arms **353L** and **353R** will be described below, the link arms **353L** and **353R** are formed to be symmetrical in the width direction W (see FIG. 2) orthogonal to the sheet feeding direction FD. Therefore, the link arm **353L** serving as a pivot member will be mainly described below with reference to FIG. 4 and a description of the link arm **353R** will be appropriately omitted.

(35) As illustrated in FIGS. 4A and 4B, the link arm **353L** includes a boss **353bL** formed at a first end portion and a boss **353cL** formed at a second end portion. The boss **353cL** is supported pivotably by the printer body **1A**, and the boss **353bL** engages with the second guide rail **350bL** of the front cover **350**. In the same manner, the link arm **353R** includes a boss **353bR** formed at a first end portion and a boss **353cR** formed at a second end portion (see FIG. 2). The boss **353cR** is supported pivotably by the printer body **1A** and the boss **353bR** engages with the second guide rail **350bR** of the front cover **350**.

(36) FIG. 5A is a perspective view illustrating the manual feed tray **351** seen from a stacking plane **351a**, i.e., an upper surface side, and FIG. 5B is a perspective view illustrating the manual feed tray **351** seen from an opposite surface, i.e., a bottom surface side of the stacking plane **351a**. The manual feed tray **351** serving as a first supporting portion is movable between a closed position, i.e., a second closed position illustrated in FIG. 7 and an open position, i.e., a second open position illustrated in FIG. 8. The manual feed tray **351** is accommodated in the printer body **1A** when the manual feed tray **351** is positioned at the closed position and is capable of supporting the sheet when the manual feed tray **351** is positioned at the open position.

(37) As illustrated in FIGS. 5A and 5B, the manual feed tray **351** includes a stacking plane **351a** formed on the upper surface of the manual feed tray **351** and on which the sheet is stacked and first shafts **351bL** and **351bR** and second shafts **351cL** and **351cR** projecting from both side surface of the manual feed tray **351**. The second shafts **351cL** and **351cR** serving as first bosses engage respectively with holes **353aL** and **353aR** formed through the bosses **353bL** and **353bR** of the link arms **353L** and **353R**.

(38) First shafts **351bL** and **351bR** serving as second bosses of the manual feed tray **351** engage respectively with the first guide rails **350aL** and **350aR** formed on the tray supporting portions **350kL** and **350kR** of the front cover **350**. The first shafts **351bL** and **351bR** are held slidably within the first guide rails **350aL** and **350aR**.

(39) Holding guides **351dL** and **351dR** are provided at an upstream end portion in the sheet feeding direction FD and at both end portions in the width direction W of the manual feed tray **351**. Still further, wall portions **351fL** and **351fR** are formed at both end portions in the width direction W of the manual feed tray **351**. An inclined portion **351e** is formed at the upstream end portion in the sheet feeding direction FD of the manual feed tray **351**.

(40) As illustrated in FIG. 6, the extension tray **352** serving as a second supporting portion is provided with a pair of guide ribs **352aL** and **352aR** at both widthwise end portions, and a grip portion **352d**, that can be gripped by the user, is provided at the upstream end portion in the sheet feeding direction FD. The pair of guide ribs **352aL** and **352aR** are held slidably by the holding guides **351dL** and **351dR** of the manual feed tray **351** (see FIGS. 5A and 5B). The more the grip

portion **352d** extends in the width direction W, the more the grip portion **352d** projects upward so as to have a function of holding the sheet stacked in the manual feed tray **351** and the extension tray **352** from not slipping out of the extension tray **352**.

(41) A pair of abutment portions **352eL** and **352eR** and a pair of claw portions **352bL** and **352bR** are provided at a downstream end portion in the sheet feeding direction FD of the extension tray **352**. The pair of claw portions **352bL** and **352bR** is disposed inside in the width direction W of the pair of abutment portions **352eL** and **352eR**. The pair of abutment portions **352eL** and **352eR** abuts with an under surface of the manual feed tray **351** in a state in which the extension tray **352** is drawn out of the manual feed tray **351**. That is, the extension tray **352** is supported by the holding guides **351dL** and **351dR** from underside and abuts with the under surface of the manual feed tray **351** so as to keep a posture of being drawn out of the manual feed tray **351**.

(42) Opening Operation of Front Cover

(43) Next, an opening operation of the front cover **350** will be described with reference to FIGS. 7 and 8. Note that because one side in the width direction W of the manual sheet feed portion **500** operates in the same manner with another side, only a left side in the width direction W of the manual sheet feed portion **500**, i.e., a side denoted by the reference sign 'L', will be described below. FIG. 7 is a section view illustrating a state in which the front cover **350** and the manual feed tray **351** are positioned at the closed position, and FIG. 8 is a section view illustrating a state in which the front cover **350** and the manual feed tray **351** are positioned at the open position.

(44) As illustrated in FIG. 7, the exterior face **350m** of the front cover **350** is disposed so as to vertically extend to configure a part of the exterior face of the printer **1** when the front cover **350** is positioned at the closed position. At this time, the manual feed tray **351**, the extension tray **352**, the link arm **353L** and the sheet feed unit **354** are accommodated inside of the printer body **1A** and take a posture of erecting approximately in the vertical direction. The manual feed tray **351** and the extension tray **352** are nonvisible from the user by being covered by the ribs **350c**, **350g** and **350h** of the front cover **350**.

(45) When the front cover **350** is positioned at the closed position, the second shaft **351cL** of the manual feed tray **351** is positioned at a side closer to an upper end portion of the second guide rail **350bL** and to the exterior face **350m**. In the same manner, the first shaft **351bL** of the manual feed tray **351** is positioned at a side closer to an upper end portion of the first guide rail **350aL** and to the exterior face **350m**. A distance between a center C1 of the second shaft **351cL** and the exterior face **350m** at this time is defined as a distance D1, and a distance between a center C2 of the first shaft **351bL** and the exterior face **350m** is defined as a distance D2. Note that the distance D1 is greater than the distance D2, i.e., $D1 > D2$, in the present exemplary embodiment. The second shaft **351cL** may be called as a first part of the manual feed tray **351**, and the first shaft **351bL** may be called as a second part of the manual feed tray **351**. The second shaft **351cL** is positioned downstream of the first shaft **351bL** in terms of the sheet feeding direction FD. The extension tray **352** is drawn out of the manual feed tray **351** in a direction opposite to the sheet feeding direction FD. Therefore, the second shaft **351cL** is positioned upstream of the first shaft **351bL** in terms of the direction in which the extension tray **352** is drawn out of the manual feed tray **351**.

(46) The front cover **350** can be opened and closed centering on an opening and closing shaft **355**, and the user can conduct an opening operation on the front cover **350** from the closed position as illustrated in FIG. 7 to the open position as illustrated in FIG. 8. As the opening operation is conducted on the front cover **350** as illustrated in FIGS. 7 and 8, the boss **353bL** of the link arm **353L** moves within the second guide rail **350bL** and abuts with the abutment surface **350nL** of the second guide rail **350bL**. Thereby, the front cover **350** is held at the open position. Still further, because the second shaft **351cL** of the manual feed tray **351** is engaged with a hole **353aL** formed through the boss **353bL** (see FIG. 4B), the second shaft **351cL** moves together with the boss **353bL** and is positioned in a vicinity of the abutment surface **350nL**.

(47) Similarly to that, as the front cover **350** is opened, the first shaft **351bL** of the manual feed tray

351 moves within the first guide rail **350aL** and abuts with an abutment surface **350pL** of the first guide rail **350aL**. These abutment surfaces **350nL** and **350pL** are positioned at downstream end portions in the sheet feeding direction FD of the second guide rail **350bL** and the first guide rail **350aL**, respectively.

(48) Thus, the manual feed tray **351** moves from the closed position to the open position in linkage with the move of the front cover **350** from the closed position to the open position. At this time, the manual feed tray **351** moves relatively to the front cover **350**. More specifically, along with the opening operation of the front cover **350**, the first shaft **351bL** and the second shaft **351cL** of the manual feed tray **351** move within the first guide rail **350aL** and the second guide rail **350bL** such that the manual feed tray **351** moves from the closed position to the open position. That is, the link arm **353L** and the first guide rail **350aL** serving as a hole portion constitute an interlocking portion **600** that moves the manual feed tray **351** from the closed position to the open position when the front cover **350** is opened from the closed position to the open position.

(49) The first guide rail **350aL** and the second guide rail **350bL** are formed such that the closer to the downstream side in the sheet feeding direction FD, the narrower their widths become. Therefore, the first shaft **351bL** and the boss **353bL** engage with the first guide rail **350aL** and the second guide rail **350bL** with almost no rattling in the state in which the front cover **350** is positioned at the open position, so that rattling of the front cover **350** is reduced.

(50) When the front cover **350** is positioned at the open position, a distance between the center C1 of the second shaft **351cL** and the exterior face **350m** is defined as a distance D3, and a distance between the center C2 of the first shaft **351bL** and the exterior face **350m** is defined as a distance D4. Note that the distance D3 is larger than the distance D4, i.e., $D3 > D4$.

(51) Then, the distances D1, D2, D3 and D4 are set such that a relationship of $D3/D1 < D4/D2$ holds in the present exemplary embodiment. The distances D1, D2, D3 and D4 are also set such that a relationship of $(D3 - D1) < (D4 - D2)$ holds in the present exemplary embodiment.

(52) Thereby, as illustrated in FIG. 8, a moving angle $\theta 2$ of the manual feed tray **351** in moving from the closed position to the open position is smaller than a moving angle $\theta 1$ of the front cover **350** in moving from the closed position to the open position, i.e., $\theta 2 < \theta 1$. In FIG. 8, a line along the exterior face **350m** of the front cover **350** positioned at the closed position is represented by a broken line F11, and a line along the exterior face **350m** of the front cover **350** positioned at the open position is represented by a broken line F12. A line along an outer side surface of the manual feed tray **351** positioned at the closed position is also represented by a broken line F21, and a line along an outer side surface of the front cover **350** positioned at the open position is represented by a broken line F22. The broken lines F11 and F21 extend approximately in the vertical direction and are in parallel with each other in the present exemplary embodiment. Note that the broken lines F11 and F21 need not be always in parallel with each other. The moving angle $\theta 1$ is an angle formed between the broken lines F11 and F12, and the moving angle $\theta 2$ is an angle formed between the broken lines F21 and F22.

(53) That is, the manual feed tray **351** takes such a posture that the upstream end portion in the sheet feeding direction FD is lifted up with respect to an inner surface of the front cover **350**. In other words, the upstream end portion in the sheet feeding direction FD of the manual feed tray **351** is positioned at a position separated from the exterior face **350m** of the front cover **350** more than the downstream end portion in the sheet feeding direction FD of the manual feed tray **351**. The extension tray **352** slidably or drawably supported by the manual feed tray **351** also takes such a posture that an upstream end portion in the sheet feeding direction FD is lifted up with respect to the inner surface of the front cover **350**. That is, when the manual feed tray **351** moves from the closed position to the open position, the upstream end portion in the sheet feeding direction FD of the manual feed tray **351** is displaced largely with respect to the front cover **350** more than the downstream end portion in the sheet feeding direction FD.

(54) Draw-Out Operation of Extension Tray

(55) Next, a draw-out operation of the extension tray **352** will be described with reference to FIGS. **9** and **10**. The user can draw out the extension tray **352** toward upstream in the sheet feeding direction FD by grasping the grip portion **352d** of the extension tray **352** in a state in which the front cover **350** is positioned at the open position. At this time, the guide rib **352aL** of the extension tray **352** is supported slidably by the holding guide **351dL** of the manual feed tray **351**.

(56) Then, as the claw portion **352b** of the extension tray **352** engages with an end portion of an inclined portion **351e** of the manual feed tray **351**, the extension tray **352** is held in a state of being drawn out of the manual feed tray **351**. The claw portion **352bL** is also provided with an inclined plane **352c**. The inclined plane **352c** protrudes in a sheet thickness direction, i.e., in a direction separating from the extension tray **352**, to close a part of a gap formed between the manual feed tray **351** and the extension tray **352**. The inclined plane **352c** also enables to smoothly deliver a sheet to the inclined portion **351e** of the manual feed tray **351**.

(57) The user often sets a sheet on the manual feed tray **351** while sliding a leading edge of the sheet along the upper surface of the extension tray **352** in the state in which the extension tray **352** is drawn out of the manual feed tray **351**. If there is a level difference between the extension tray **352** and the manual feed tray **351** at this time, there is a possibility that the leading edge of the sheet is caught by the level difference and sheet settability is hampered. However, because the sheet is smoothly guided from the inclined plane **352c** of the claw portion **352bL** to the inclined portion **351e** of the manual feed tray **351** in the present exemplary embodiment, the sheet settability can be improved.

(58) By the way, the front cover **350** is required to have a certain degree of stiffness such that the front cover **350** is not bent when the user operates to open or close the front cover **350**. To that end, the front cover **350** includes the rib **350c** at the upstream end portion in the sheet feeding direction FD thereof as described above. Still further, downsizing of the printer **1** and the manual sheet feed portion **500** is demanded lately. That is, it is desired to assure stiffness of the front cover **350** while realizing downsizing of the manual sheet feed portion **500** in a state in which the front cover **350** is closed.

(59) For instance, in a case where the rib **350c** is provided at the upstream end portion in the sheet feeding direction FD of the front cover **350** as described in the present exemplary embodiment, it is necessary to draw out the extension tray **352** while avoiding the rib **350c**. In a case where the draw-out direction of the extension tray **352** is approximately in parallel with the exterior face **350m** of the front cover **350**, the extension tray **352** has to be disposed at a height that exceeds the rib **350c**, thus enlarging the manual sheet feed portion **500**.

(60) Then, according to the present exemplary embodiment, the manual feed tray **351** is accommodated in a posture of running along the front cover **350** in the state in which the front cover **350** is positioned at the closed position to downsize the manual sheet feed portion **500** as illustrated in FIG. **7**. When the extension tray **352** is seen along the draw-out direction of the extension tray **352** in this state, the extension tray **352** overlaps with the rib **350c**. Meanwhile, in the state in which the front cover **350** is positioned at the open position as illustrated in FIG. **8**, the manual feed tray **351** takes a posture in which the upstream end portion in the sheet feeding direction FD is lifted up with respect to the front cover **350**. In this state, the extension tray **352** does not overlap with the rib **350c** when the extension tray **352** is seen along the draw-out direction. The manual feed tray **351** takes the position in which the upstream end portion in the sheet feeding direction FD is lifted up from the front cover **350**. Then, the holding guide **351dL** of the manual feed tray **351** guides the draw-out operation of the extension tray **352** such that the extension tray **352** does not abut with the rib **350c**. Therefore, the extension tray **352** can be drawn out of the manual feed tray **351** such that the extension tray **352** passes above the rib **350c** without interfering with the rib **350c**.

(61) The manual feed tray **351** automatically takes a position as illustrated in FIG. **8** by opening the front cover **350**, and the extension tray **352**, slidably supported by the manual feed tray **351**,

follows the manual feed tray **351**. Then, because the user performs a drawing-out operation of the extension tray **352**, mechanical reliability is high and it is possible to achieve both improvements of stiffness of the front cover **350** and of drawability of the extension tray **352**. Still further, because the manual feed tray **351** does not take such a posture that the upstream end portion in the sheet feeding direction FD is lifted up with respect to the front cover **350** in the state in which the front cover **350** is positioned at the closed position, the apparatus can be downsized.

(62) The position of the extension tray **352** is restricted widthwise by the wall portions **351fL** and **351fR** of the manual feed tray **351** and vertically by the restriction ribs **350dL** and **350dR**.

Therefore, it is possible to reduce rattling during the drawing operation of the extension tray **352** and to improve drawing operability of the extension tray **352**.

(63) It is also conceivable that the user operates and closes the front cover **350** in a state in which the extension tray **352** is drawn out by forgetting to accommodate the extension tray **352** into the manual feed tray **351**. When the front cover **350** is closed from the open position to the closed position, the upstream end portions in the sheet feeding direction FD of the manual feed tray **351** and the extension tray **352** approach to the inner surface of the front cover **350**. However, because the extension tray **352** interferes with the rib **350c** of the front cover **350** in the state in which the extension tray **352** is drawn out, there is a possibility that the manual feed tray **351** and the extension tray **352** are damaged if the user tries to forcibly close the front cover **350**.

(64) Then, according to the present exemplary embodiment, the first guide rail **350aL** of the front cover **350** is formed such that the closer to an outside in a radial direction orthogonal to the pivot direction of the front cover **350**, the wider the width of the first guide rail **350aL** is as illustrated in FIG. 10. Therefore, the first shaft **351bL** is movable by deviating from a moving locus of the first shaft **351bL** in opening or closing the front cover **350** in the state in which the extension tray **352** is accommodated in the manual feed tray **351** as illustrated in FIGS. 7 and 8.

(65) If the user tries to close the front cover **350** from the open position to the closed position in the state in which the extension tray **352** is drawn out, the first shaft **351bL** of the manual feed tray **351** moves from the position as illustrated in FIG. 8 in a direction of gradually approaching to the front side of the front cover **350** and to the exterior face **350m** of the front cover **350**. However, the extension tray **352** being drawn out interferes with the rib **350c** of the front cover **350** on a way of the closing operation of the front cover **350**.

(66) If the user tries to close the front cover **350** further to the closed position at this time, the first shaft **351bL** moves to an upper left corner of the first guide rail **350aL** as illustrated in FIG. 10.

That is, the first shaft **351bL** moves within the first guide rail **350aL** so as to release a force received by the extension tray **352** from the printer body **1A**. Due to that, it is possible to mitigate impact received by the manual feed tray **351** and the extension tray **352** from the printer body **1A** and to suppress the manual feed tray **351** and the extension tray **352** from being damaged.

(67) As described above, according to the present exemplary embodiment, the front cover **350** is provided with the ribs **350c**, **350g** and **350h** to improve stiffness of the front cover **350**. The rib **350c** is positioned at the position farther from the opening and closing shaft **355** serving as an opening and closing center of the front cover **350** than the extension tray **352** being attached to the manual feed tray **351** as illustrated in FIG. 7. Space saving of the manual sheet feed portion **500** is realized when the front cover **350** is positioned at the closed position and the sheet feeding apparatus can be downsized by the manual feed tray **351**, the extension tray **352** and the rib **350c** constructed as described above.

(68) Still further, the moving angle θ_2 of the manual feed tray **351** in moving from the closed position to the open position is smaller than the moving angle θ_1 of the front cover **350** in moving from the closed position to the open position as illustrated in FIG. 8. The upstream end in the sheet feeding direction FD of the manual feed tray **351** is lifted up from the front cover **350** by opening the front cover **350** to the open position by thus setting the opening angles of the manual feed tray **351** and the front cover **350**.

(69) This arrangement makes it possible to smoothly draw out the extension tray **352** out of the manual feed tray **351** without interfering with the rib **350c** of the front cover **350** and to improve drawing operability of the extension tray **352**. Thus, it is possible to improve the drawing operability of the extension tray **352** while increasing the stiffness of the front cover **350** and downsizing the sheet feeding apparatus.

Other Embodiments

(70) It is noted that while the front cover **350** is provided with the ribs **350c**, **350g** and **350h** in the exemplary embodiment described above, the present disclosure is not limited to such configuration. For instance, it is not necessary to provide all of the ribs **350c**, **350g** and **350h** on the front cover **350**, and either one or two of the ribs **350c**, **350g** and **350h** may be provided on the front cover **350**. Still further, shapes, sizes and dispositions of the ribs provided on the front cover **350** are not limited.

(71) The manual feed tray **351** includes the second shafts **351cL** and **351cR** supported by the link arms **353L** and **353R** and the first shafts **351bL** and **351bR** supported by the first guide rails **350aL** and **350aR** of the front cover **350** in the exemplary embodiment described above. Then, while the manual feed tray **351** is opened or closed by moving the second shafts **351cL** and **351cR** and the first shafts **351bL** and **351bR** in opening or closing the front cover **350**, the present disclosure is not limited to such configuration. That is, the configuration of the interlocking mechanism of the front cover **350** and the manual feed tray **351** is not limited such configuration.

(72) Still further, while the electro-photographic printer **1** has been used to describe the exemplary embodiment described above, the present disclosure is not limited to such a printer. For instance, the present disclosure is also applicable to an inkjet type image forming apparatus configured to form an image on a sheet by discharging ink droplets from a nozzle. The present disclosure may be applied also to a sheet feeding apparatus of an optional sheet feeding deck connected to an image forming apparatus.

(73) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(74) This application claims the benefit of Japanese Patent Application No. 2020-205494, filed Dec. 11, 2020, which is hereby incorporated by reference herein in its entirety.

Claims

1. A sheet feeding apparatus comprising: an apparatus body; a cover openably and closably supported between a first closed position and a first open position with respect to the apparatus body, the cover including a first surface; a first tray movable between a second closed position where the first tray is accommodated in the apparatus body and a second open position where the first tray is capable of supporting a sheet, the first tray including a second surface; an interlocking portion configured to move the first tray from the second closed position to the second open position in a case where the cover is opened from the first closed position to the first open position; a second tray drawably supported by the first tray and capable of supporting the sheet together with the first tray; and a feed portion configured to feed the sheet supported by the first and second trays in a sheet feeding direction, wherein when the cover moves from the first closed position to the first open position, the second tray moves relative to the cover so that the second tray is lifted up from the cover, wherein when the cover moves from the first closed position to the first open position, the cover moves so that a first straight line along the first surface moves from a first position to a second position, and when the first tray moves from the second closed position to the second open position, the first tray moves so that a second straight line along the second surface moves from a third position to a fourth position, and wherein a narrow angle between the second straight line at

the third position and the second straight line at the fourth position is smaller than a narrow angle between the first straight line at the first position and the first straight line at the second position, and wherein the first tray includes a guide configured to guide the second tray while the second tray is being drawn out of the first tray.

2. The sheet feeding apparatus according to claim 1, wherein the second tray is capable of being drawn out of the first tray without interfering with the cover in a case where the cover is positioned at the first open position and the first tray is positioned at the second open position.

3. The sheet feeding apparatus according to claim 2, wherein the cover comprises a rib positioned at a position farther from an opening and closing center of the cover than the second tray being attached to the first tray, and wherein the second tray is capable of being drawn out of the first tray so as to pass above the rib without interfering with the rib in a case where the cover is positioned at the first open position and the first tray is positioned at the second open position.

4. The sheet feeding apparatus according to claim 1, wherein the interlocking portion comprises a pivot member pivotably supported by the apparatus body and a hole portion provided through the cover, wherein the first tray comprises a first boss supported by the pivot member and a second boss supported by the hole portion, and wherein a relationship of $D3/D1 < D4/D2$ holds in a case where a distance between a center of the first boss and an exterior face of the cover and a distance between a center of the second boss and the exterior face are defined respectively as D1 and D2 when the cover is positioned at the first closed position and the first tray is positioned at the second closed position, and in a case where a distance between the center of the first boss and the exterior face and a distance between the center of the second boss and the exterior face are defined respectively as D3 and D4 when the cover is positioned at the first open position and the first tray is positioned at the second open position.

5. The sheet feeding apparatus according to claim 4, wherein the second boss is movable within the hole portion so as to release a force received by the second tray from the apparatus body in a case where the cover is closed from the first open position to the first closed position while drawing the second tray out of the first tray.

6. The sheet feeding apparatus according to claim 3, wherein the rib extends in a direction of separating from an exterior face of the cover.

7. The sheet feeding apparatus according to claim 3, wherein the rib extends in a width direction orthogonal to the sheet feeding direction.

8. An image forming apparatus comprising: the sheet feeding apparatus according to claim 1; and an image forming unit configured to form an image on a sheet fed from the sheet feeding apparatus.

9. The sheet feeding apparatus according to claim 1, wherein the first tray includes an upstream end portion and a downstream end portion with respect to the sheet feeding direction, and wherein in a state where the cover is in the first open position and the first tray is in the second open position, the upstream end portion of the first tray is further from an exterior surface of the cover than the downstream end portion of the first tray.

10. The sheet feeding apparatus according to claim 1, wherein the interlocking portion comprises a first pivot member pivotably supported by the apparatus body, wherein the first pivot member includes a first engaging portion, wherein the cover is provided with a first hole and a second hole, wherein the first tray includes a first boss and a second boss, wherein at least one of the first engaging portion and the first boss is inserted into the first hole, and wherein the second boss is inserted into the second hole.

11. The sheet feeding apparatus according to claim 10, wherein the cover is pivotable about an axis so that the cover moves between the first closed position and the first open position, and wherein the first hole is closer to the axis than the second hole.

12. The sheet feeding apparatus according to claim 10, wherein the interlocking portion comprises a second pivot member pivotably supported by the apparatus body, wherein the second pivot member includes a second engaging portion, wherein the cover is provided with a third hole and a

fourth hole, wherein the first tray includes a third boss and a fourth boss, wherein at least one of the second engaging portion and the third boss is inserted into the third hole, and wherein the fourth boss is inserted into the fourth hole.

13. The sheet feeding apparatus according to claim 12, wherein the cover is pivotable about an axis so that the cover moves between the first closed position and the first open position, wherein the first hole is closer to the axis than the second hole, and wherein the third hole is closer to the axis than the fourth hole.

14. The sheet feeding apparatus according to claim 1, wherein the second tray is configured to contact the first tray so as to keep a posture of the second tray drawn out from the first tray.

15. A sheet feeding apparatus comprising: an apparatus body; a cover openably and closably supported between a first closed position and a first open position with respect to the apparatus body, the cover including a first surface; a first tray movable between a second closed position where the first tray is accommodated in the apparatus body and a second open position where the first tray is capable of supporting a sheet, the first tray including a second surface; an interlocking portion configured to move the first tray from the second closed position to the second open position in a case where the cover is opened from the first closed position to the first open position; a second tray drawably supported by the first tray and capable of supporting the sheet together with the first tray; and a feed portion configured to feed the sheet supported by the first and second trays in a sheet feeding direction, wherein when the cover moves from the first closed position to the first open position, the second tray moves relative to the cover so that the second tray is lifted up from the cover, wherein when the cover moves from the first closed position to the first open position, the cover moves so that a first straight line along the first surface moves from a first position to a second position, and when the first tray moves from the second closed position to the second open position, the first tray moves so that a second straight line along the second surface moves from a third position to a fourth position, wherein a narrow angle between the second straight line at the third position and the second straight line at the fourth position is smaller than a narrow angle between the first straight line at the first position and the first straight line at the second position, and wherein the second tray is capable of being drawn out of the first tray without interfering with the cover in a case where the cover is positioned at the first open position and the first tray is positioned at the second open position.

16. The sheet feeding apparatus according to claim 15, wherein the cover comprises a rib positioned at a position farther from an opening and closing center of the cover than the second tray being attached to the first tray, and wherein the second tray is capable of being drawn out of the first tray so as to pass above the rib without interfering with the rib in a case where the cover is positioned at the first open position and the first tray is positioned at the second open position.

17. A sheet feeding apparatus comprising: an apparatus body; a cover openably and closably supported between a first closed position and a first open position with respect to the apparatus body, the cover including a first surface; a first tray movable between a second closed position where the first tray is accommodated in the apparatus body and a second open position where the first tray is capable of supporting a sheet, the first tray including a second surface; an interlocking portion configured to move the first tray from the second closed position to the second open position in a case where the cover is opened from the first closed position to the first open position; a second tray drawably supported by the first tray and capable of supporting the sheet together with the first tray; and a feed portion configured to feed the sheet supported by the first and second trays in a sheet feeding direction, wherein when the cover moves from the first closed position to the first open position, the second tray moves relative to the cover so that the second tray is lifted up from the cover, wherein when the cover moves from the first closed position to the first open position, the cover moves so that a first straight line along the first surface moves from a first position to a second position, and when the first tray moves from the second closed position to the second open position, the first tray moves so that a second straight line along the second surface moves from a

third position to a fourth position, wherein a narrow angle between the second straight line at the third position and the second straight line at the fourth position is smaller than a narrow angle between the first straight line at the first position and the first straight line at the second position, wherein the interlocking portion comprises a first pivot member pivotably supported by the apparatus body, wherein the first pivot member includes a first engaging portion, wherein the cover is provided with a first hole and a second hole, wherein the first tray includes a first boss and a second boss, wherein at least one of the first engaging portion and the first boss is inserted into the first hole, and wherein the second boss is inserted into the second hole.

18. The sheet feeding apparatus according to claim 17, wherein the cover is pivotable about an axis so that the cover moves between the first closed position and the first open position, and wherein the first hole is closer to the axis than the second hole.

19. The sheet feeding apparatus according to claim 17, wherein the interlocking portion comprises a second pivot member pivotably supported by the apparatus body, wherein the second pivot member includes a second engaging portion, wherein the cover is provided with a third hole and a fourth hole, wherein the first tray includes a third boss and a fourth boss, wherein at least one of the second engaging portion and the third boss is inserted into the third hole, and wherein the fourth boss is inserted into the fourth hole.

20. The sheet feeding apparatus according to claim 19, wherein the cover is pivotable about an axis so that the cover moves between the first closed position and the first open position, wherein the first hole is closer to the axis than the second hole, and wherein the third hole is closer to the axis than the fourth hole.
